

**SEMESTER S6**  
**ENTREPRENEURSHIP**

|                                        |                 |                    |                |
|----------------------------------------|-----------------|--------------------|----------------|
| <b>Course Code</b>                     | <b>OEEVT614</b> | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week (L: T:P: R)</b> | 3:0:0:0         | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                         | 3               | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>          | None            | <b>Course Type</b> | Theory         |

**Course Objectives:**

1. To understand the importance of entrepreneurship and apply it in an organization.

**SYLLABUS**

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Contact Hours</b> |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | Entrepreneurship: definition, requirements to be an entrepreneur, entrepreneur and intrapreneur, entrepreneur and manager, growth of entrepreneurship in India, women entrepreneurship, rural and urban entrepreneurship.<br><br>Entrepreneurial Motivation: motivating factors, motivation theories-Maslow's need hierarchy theory, McClelland's acquired need theory, government's policy actions towards entrepreneurial motivation, entrepreneurship development programmes.                                                              | <b>9</b>             |
| <b>2</b>          | Types of Enterprises and Ownership Structure: small scale, medium scale and large scale enterprises, role of small enterprises in economic development; proprietorship, partnership, limited companies and co-operatives: their formation, capital structure and source of finance.<br><br>Institutional Support and Policies: institutional support towards the development of entrepreneurship in India, technical consultancy organizations, Government programs, policies, incentive and institutional networking for enterprise setting. | <b>9</b>             |
| <b>3</b>          | Projects: identification and selection of projects, project report, contents and formulation, elements of project formulation, project design and network analysis, concept of project evaluation, methods of project evaluation: internal                                                                                                                                                                                                                                                                                                    | <b>9</b>             |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|          | rate of return method and net present value method.                                                                                                                                                                                                                                                                                                                                                                                      |          |
| <b>4</b> | Management of Enterprises: objectives and functions of management, scientific management, general and strategic management; introduction to human resource management: planning, job analysis, training, recruitment and selection, marketing and organizational dimension of enterprises; enterprise financing, raising and managing capital, shares, debentures, bonds, cost of capital; break- even analysis, balance sheet analysis. | <b>9</b> |

**Course Assessment Method**  
(CIE: 40 marks, ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Assignment/<br/>Microproject</b> | <b>Internal<br/>Examination-1<br/>(Written)</b> | <b>Internal<br/>Examination- 2<br/>(Written )</b> | <b>Total</b> |
|-------------------|-------------------------------------|-------------------------------------------------|---------------------------------------------------|--------------|
| <b>5</b>          | <b>15</b>                           | <b>10</b>                                       | <b>10</b>                                         | <b>40</b>    |

**End Semester Examination Marks (ESE)**

*In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions*

| <b>Part A</b>                                                                                                                                                                                     | <b>Part B</b>                                                                                                                                                                                                                                                                                                      | <b>Total</b> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <ul style="list-style-type: none"> <li>• 2 Questions from each module.</li> <li>• Total of 8 Questions, each carrying 3 marks</li> </ul> <p style="text-align: center;"><b>(8x3 =24marks)</b></p> | <ul style="list-style-type: none"> <li>• Each question carries 9 marks.</li> <li>• Two questions will be given from each module, out of which 1 question should be answered.</li> <li>• Each question can have a maximum of 3 sub divisions.</li> </ul> <p style="text-align: center;"><b>(4x9 = 36 marks)</b></p> | <b>60</b>    |

### Course Outcomes (COs)

At the end of the course students should be able to:

| Course Outcome |                                                               | Bloom's Knowledge Level (KL) |
|----------------|---------------------------------------------------------------|------------------------------|
| <b>CO1</b>     | Describe the fundamental concepts of entrepreneurship         | K1, K2, K3                   |
| <b>CO2</b>     | Understand entrepreneurial motivation and motivation theories | K1, K2, K3                   |
| <b>CO3</b>     | Analyze types of enterprises and ownership structure          | K1, K2, K3                   |
| <b>CO4</b>     | Apply project evaluation methods                              | K1, K2, K3, K4, K5           |
| <b>CO5</b>     | Evaluate enterprise financial strength                        | K1, K2, K3, K4, K5           |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

### CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> |     |     |     |     |     |     |     |     |     | 1    | 1    |      |
| <b>CO2</b> |     |     |     |     |     |     |     | 2   |     | 1    |      |      |
| <b>CO3</b> |     |     |     |     |     |     |     |     |     | 1    |      |      |
| <b>CO4</b> | 2   |     | 1   |     | 2   |     |     |     |     | 1    | 3    |      |
| <b>CO5</b> | 2   |     | 1   |     | 2   |     |     |     |     | 1    | 3    |      |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| Text Books |                              |                      |                                  |                  |
|------------|------------------------------|----------------------|----------------------------------|------------------|
| Sl. No     | Title of the Book            | Name of the Author/s | Name of the Publisher            | Edition and Year |
| 1          | Entrepreneurial Development, | Khanka,SS            | S Chand & Company Ltd. New Delhi | 2007             |
| 2          | Entrepreneurial Development  | Ram Chandran         | Tata McGraw Hill, New Delhi      | 2008             |

| Reference Books |                                                      |                      |                              |                  |
|-----------------|------------------------------------------------------|----------------------|------------------------------|------------------|
| Sl. No          | Title of the Book                                    | Name of the Author/s | Name of the Publisher        | Edition and Year |
| 1               | Entrepreneurial Development Programmes and Practices | Saini, J. S          | Deep & Deep Publications     | ,2012            |
| 2               | Entrepreneurship for Engineers                       | Badhai, B            | Dhanpat Rai & co             | 2006             |
| 3               | Project Management and Entrepreneurship              | Desai, Vasant        | Himalayan Publishing, Mumbai | 2017             |
| 4               | Entrepreneurial Development                          | Gupta, Srinivasan    | S Chand & Sons, New Delhi    | 2020             |

| Video Links (NPTEL, SWAYAM...) |                                                                                           |
|--------------------------------|-------------------------------------------------------------------------------------------|
| Module No.                     | Link ID                                                                                   |
| 1                              | <a href="https://nptel.ac.in/courses/110106141">https://nptel.ac.in/courses/110106141</a> |
| 2                              | <a href="https://nptel.ac.in/courses/110106141">https://nptel.ac.in/courses/110106141</a> |
| 3                              | <a href="https://nptel.ac.in/courses/110106141">https://nptel.ac.in/courses/110106141</a> |
| 4                              | <a href="https://nptel.ac.in/courses/110106141">https://nptel.ac.in/courses/110106141</a> |

**SEMESTER 6**

**BIOMEDICAL ENGINEERING**

|                                            |                  |                    |                |
|--------------------------------------------|------------------|--------------------|----------------|
| <b>Course Code</b>                         | <b>OEETVT615</b> | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 3:0:0:0          | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                             | 3                | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>              | None             | <b>Course Type</b> | Theory         |

**Course Objectives:**

1. This course will introduce aspects of biomedical engineering as applied to biological systems described using engineering principles and the use of modern diagnostic and therapeutic equipment.

**SYLLABUS**

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                             | <b>Contact Hours</b> |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | <b>Physiology and Bio- Electric Concepts</b><br>Cell and their structure, Transport of ions through the cell membrane, Resting and action potentials, Bio-electric potentials.<br>Electrode theory – Electrode potential, Types of electrodes, Equivalent circuit.<br>Instrumentation amplifier, Isolation amplifier, Chopper amplifier, Devices to protect against electrical hazards. | <b>9</b>             |
| <b>2</b>          | <b>Body Potential Measurement</b><br>ECG – Nature of ECG waveform, ECG lead configurations, ECG recorder.<br>EEG – Evoked potentials, Brain waves, Analysis of EEG<br>EMG – Recording setup<br>ERG and EOG                                                                                                                                                                              | <b>9</b>             |
| <b>3</b>          | <b>Prosthesis</b><br>Heart Lung machine – Model of the heart-lung machine.<br>Kidney machine – Dialysis.                                                                                                                                                                                                                                                                                | <b>9</b>             |

|          |                                                                                                                                                                                                                                                                              |          |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
|          | Nerve stimulators – Diagnostic/ Therapeutic stimulator.<br>Centralized & Bedside monitoring<br>Microprocessor based ventilator                                                                                                                                               |          |
| <b>4</b> | <b>Medical Imaging</b><br>Computer Tomography – Basic principle, Image construction, Block diagram, Applications.<br>Magnetic Resonance Imaging – Block diagram, Image reconstruction.<br>Ultrasonic Imaging – Different modes.<br>Positron Emission Tomography – Principle. | <b>9</b> |

**Course Assessment Method**  
(CIE: 40 marks, ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Assignment/<br/>Microproject</b> | <b>Internal<br/>Examination-1<br/>(Written)</b> | <b>Internal<br/>Examination- 2<br/>(Written )</b> | <b>Total</b> |
|-------------------|-------------------------------------|-------------------------------------------------|---------------------------------------------------|--------------|
| <b>5</b>          | <b>15</b>                           | <b>10</b>                                       | <b>10</b>                                         | <b>40</b>    |

**End Semester Examination Marks (ESE)**

*In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions*

| <b>Part A</b>                                                                                                                                                                                     | <b>Part B</b>                                                                                                                                                                                                                                                                                                      | <b>Total</b> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <ul style="list-style-type: none"> <li>• 2 Questions from each module.</li> <li>• Total of 8 Questions, each carrying 3 marks</li> </ul> <p style="text-align: center;"><b>(8x3 =24marks)</b></p> | <ul style="list-style-type: none"> <li>• Each question carries 9 marks.</li> <li>• Two questions will be given from each module, out of which 1 question should be answered.</li> <li>• Each question can have a maximum of 3 sub divisions.</li> </ul> <p style="text-align: center;"><b>(4x9 = 36 marks)</b></p> | <b>60</b>    |

### Course Outcomes (COs)

At the end of the course students should be able to:

| Course Outcome |                                                              | Bloom's Knowledge Level (KL) |
|----------------|--------------------------------------------------------------|------------------------------|
| CO1            | Learn the basics of bioelectric potentials                   | K1                           |
| CO2            | Understand the operation of different biopotential recorders | K2                           |
| CO3            | Learn the medical equipment for diagnosis and therapy        | K1                           |
| CO4            | Understand the general concepts of imaging systems           | K2                           |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

### CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

|     | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| CO1 | 3   | 2   |     |     |     |     |     |     |     |      |      | 2    |
| CO2 | 3   | 2   |     |     |     |     |     |     |     |      |      | 2    |
| CO3 | 3   | 3   |     |     |     |     |     |     |     |      |      | 2    |
| CO4 | 3   | 3   |     |     |     |     |     |     |     |      |      | 2    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| Text Books |                                             |                                                   |                       |                  |
|------------|---------------------------------------------|---------------------------------------------------|-----------------------|------------------|
| Sl. No     | Title of the Book                           | Name of the Author/s                              | Name of the Publisher | Edition and Year |
| 1          | Handbook of Biomedical Instrumentation      | R S Khandpur                                      | TATA McGRAW HILL      |                  |
| 2          | Biomedical Instrumentation and Measurements | Leslie Cromwell, Fred J Weibell, Erich A Pfeiffer | PHI                   |                  |

| Reference Books |                                                    |                                |                       |                  |
|-----------------|----------------------------------------------------|--------------------------------|-----------------------|------------------|
| Sl. No          | Title of the Book                                  | Name of the Author/s           | Name of the Publisher | Edition and Year |
| 1               | Medical Instrumentation<br>Application and Design  | John G Webster                 | WILEY                 |                  |
| 2               | Introduction to Biomedical<br>Equipment Technology | Joseph J Carr, John M<br>Brown | PEARSON               |                  |

| Video Links (NPTEL, SWAYAM...) |                                                                                                                             |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Module No.                     | Link ID                                                                                                                     |
| 1                              | <a href="https://www.youtube.com/watch?v=OqNDFF1RsMU">https://www.youtube.com/watch?v=OqNDFF1RsMU</a>                       |
| 2                              | <a href="https://www.youtube.com/watch?v=mK6sPBbChqc">https://www.youtube.com/watch?v=mK6sPBbChqc</a>                       |
| 3                              | <a href="https://onlinecourses.swayam2.ac.in/nou23_bt05/preview">https://onlinecourses.swayam2.ac.in/nou23_bt05/preview</a> |
| 4                              | <a href="https://onlinecourses.nptel.ac.in/noc22_bt56/preview">https://onlinecourses.nptel.ac.in/noc22_bt56/preview</a>     |